1. An object is 40 cm in front of a concave mirror with a focal length of 20 cm. Use ray tracing to locate the image. Is the image upright or inverted?

2. An object is 12 cm in front of a concave mirror with a focal length of 20 cm. Use ray tracing to locate the image. Is the image upright or inverted?

3. A 3.0-cm-tall object is 20 cm in front of a convex mirror with a radius of curvature of 120 cm. Use ray tracing to locate the image and then calculate the position and height of the image. and the magnification. State if the image is in front or behind the mirror, and whether it is upright or inverted.

4. A 2.0-cm-tall object is placed in front of a mirror. A 1.0-cm-tall upright image is formed behind the mirror, 150 cm from the object. What is the focal length of the mirror?

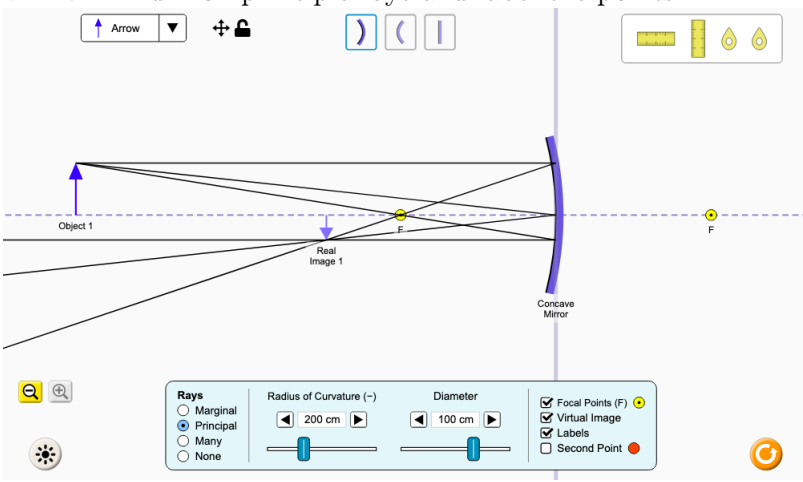
5. A convex mirror with a focal length of -75 cm is used to give a truck driver a view of the vehicle.

   A. If a person who is 1.7 m tall stands 2.2 m from the mirror, where is the person's image located?

   B. Is the image upright or inverted?

   C. What is the size of the image?

6. Go to: `https://phet.colorado.edu/sims/html/geometric-optics/latest/geometric-optics_en.html`. Turn on principle rays and label the points



   A. For a concave mirror with |R|= 150 cm, where must the object be, so that the image is in the same spot?

   B. For a concave mirror with |R|= 200 cm, where must the object be, so that the image switches from being real to virtual?

   C. For a convex mirror with |R|= 200, is it possible to get a real image?

   D. Show that for a flat mirror $d_o = d_i$.

   E. Estimate the magnification for a concave mirror with |R| = 200, when $d_0 = 3f$.